Page No. :
Date :

original

Title :- Finding minimum Travel Times From a selected Bus stop using Dijkshtra's algorithm

problem statement
Represent bus stops as a vertices in weighted graph and bus routes and Edges. Each Edge weight represent Travel time. Give a selected source stop. apply Dijkstra's algorithm to calculate and display The minimum Travel time from That source to every reachable stop. stop That cannot be reached must be identified clearly.

Solution
Represent The Network with an adjancy list. initialize The source distance To zero and all other distance to infinity. use a min-prority queue to repeadtly process The stop with The smallest known distance. for every connected task. stop. ~~td~~ update its distance if a shorter travel time is found. Finally print The minimum Travel time for each stop or display.

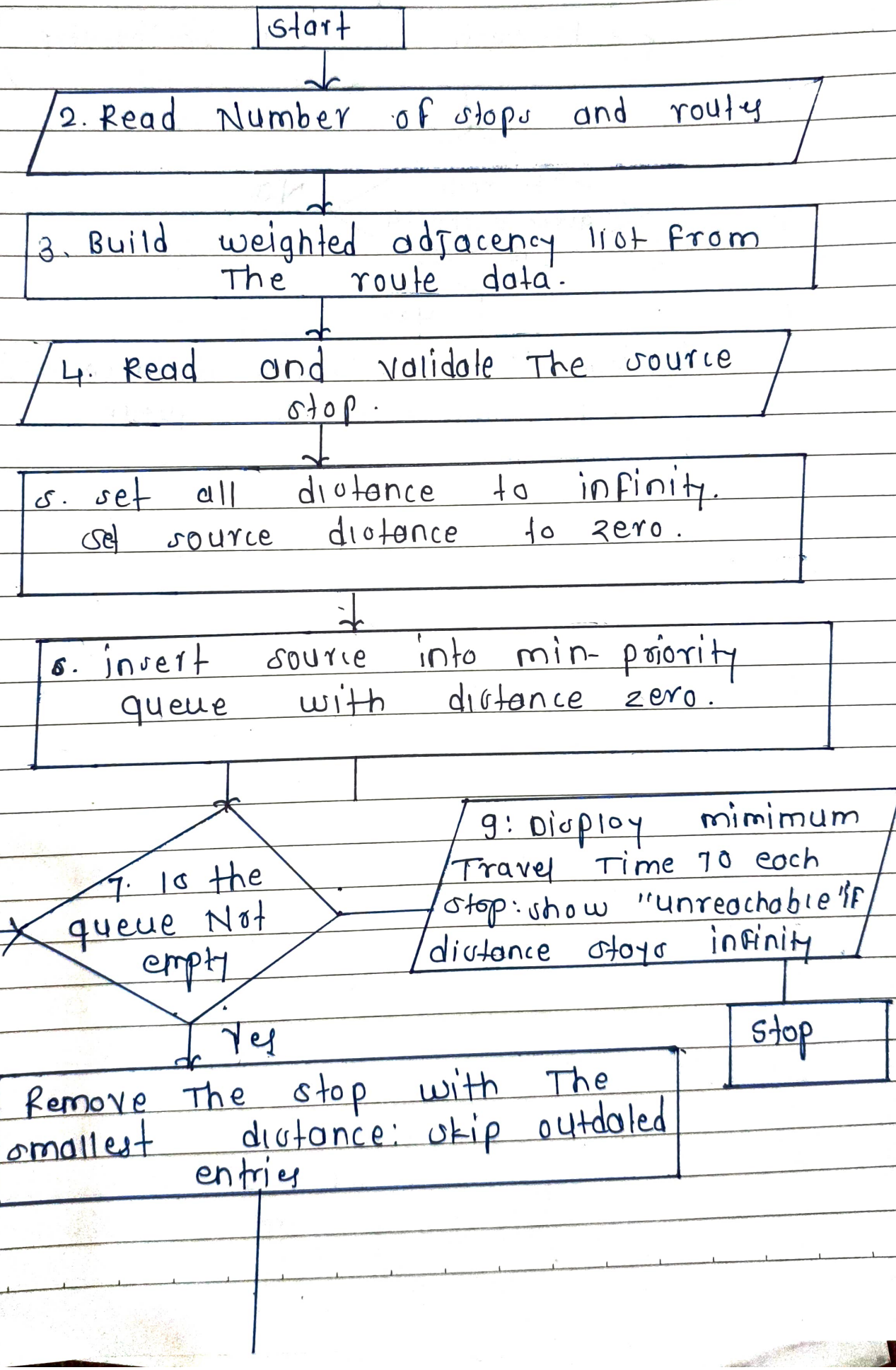
Date :

Algorithm

1. start
2. Read The Number of stop and route.
3. Build a weighted adjency list from The route data
4. Read The and validate The source stop.
5. set all distances To infinity and set source distance to zero.
6. insert The source into a min-priority queue with distance to zero.
7. while The queue is not empty. remove the stop with The smallest distance. skip outdated entries.
8. For each Neighbor. if The current routes gives a shorter distance. update The Neighbor's distant and insert it into the queue.
9. Display The minimum Travel time to each stop display "unreachable". For stop whose distance remains infinity.

Page No. :
Date :

Flow chart



unit-5

Page No. :
Date :

